

Unit 5 Pre-Test

The Pythagorean Theorem

Standards assessed: NC.8.G.6, NC.8.G.7, NC.8.G.8

Name: _____ Date: _____ Period: _____

SCORE **3.0** Basic Understanding

NC.8.G.7 Apply the Pythagorean theorem to find the hypotenuse.

• I can use $a^2 + b^2 = c^2$ to find the hypotenuse of a right triangle.

Match each vocabulary term to its definition, then find each hypotenuse.

– ▼ hypotenuse	A. The two sides that form the right angle.
– ▼ legs	B. $a^2 + b^2 = c^2$, relating the sides of a right triangle.
– ▼ Pythagorean theorem	C. The side opposite the right angle; the longest side.
– ▼ right triangle	D. A triangle that contains one 90° angle.
– ▼ square root	E. A number that, multiplied by itself, gives a given value.
– ▼ converse	F. If $a^2 + b^2 = c^2$, then the triangle is a right triangle.

Pythagorean theorem — reference

$$a^2 + b^2 = c^2$$

c is the hypotenuse (opposite the right angle); **a** and **b** are the legs.

1. Find *c*

Legs 3 and 4. Find the hypotenuse.

c =

2. Find *c*

Legs 6 and 8. Find the hypotenuse.

c =

3. Find c

Legs 5 and 12. Find the hypotenuse.

$c =$

5. Find c

Legs 9 and 12. Find the hypotenuse.

$c =$

4. Find c

Legs 8 and 15. Find the hypotenuse.

$c =$

6. Find c

Legs 7 and 24. Find the hypotenuse.

$c =$

SCORE 4.0 Target Content

NC.8.G.7 Find a missing leg and estimate non-perfect square roots.

• I can find a missing leg and round answers to the nearest tenth when needed.

Find each missing side. Round to the nearest tenth where asked.

7. Find a leg

One leg 6, hypotenuse 10. Find the other leg.

$a =$

8. Find a leg

One leg 5, hypotenuse 13. Find the other leg.

$a =$

9. Find a leg

One leg 9, hypotenuse 15. Find the other leg.

$a =$

10. Nearest tenth

Legs 2 and 3. Find the hypotenuse to the nearest tenth.

$c \approx$

11. Nearest tenth

12. Nearest tenth

Legs 5 and 7. Find the hypotenuse to the nearest tenth.

$c \approx$

One leg 4, hypotenuse 9. Find the other leg to the nearest tenth.

$a \approx$

SCORE **5.0** Advanced Content

NC.8.G.8 Apply the theorem to the coordinate plane and real-world problems.

• *I can find the distance between two points and solve real-world right-triangle problems.*

Solve each problem.

13. Distance

Find the distance between (0, 0) and (6, 8).

$d =$

14. Distance

Find the distance between (1, 2) and (4, 6).

$d =$

15. Apply

A 13 ft ladder reaches 12 ft up a wall. How far is the base from the wall?

$=$

16. Apply

A rectangle is 9 by 12. Find the length of its diagonal.

$d =$

17. Distance

Find the distance between (-2, -1) and (3, 11).

$d =$

18. Apply

A park is a right triangle with legs 20 m and 21 m. Find the longest side.

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